11.4 The Tautochrone

WORKSHEET

The question

STEP III 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q9

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The bowl is smooth. Which forces act on the particle, and which of them does work?

STAGE 1 Energy along the curve

Covers: the constant

Ask yourself The bowl is smooth. Which forces act, and which of them does work? The speed along the surface is $|\frac{ds}{dt}|$. Write down conservation of energy.

STAGE 2 Simple harmonic motion in s

Covers: the time to reach V

Pause and try Use $s = 2\sqrt{ky}$ to replace y by a function of s. Differentiate the energy equation with respect to t. What kind of motion do you get?

STAGE 3 The time to the bottom

Covers: the time to reach V

Ask yourself What fraction of a period does the particle take to reach $s = 0$? Does the answer depend on h?

STAGE 4 Two particles

Covers: the two particles

Ask yourself When does each particle reach V? How does the speed at V depend on the starting height? Compare the momenta.

Pause and try The total momentum at the collision is zero, so it stays zero. Use the coefficient of restitution α to find the two speeds just after the collision.

STAGE 5 **The whole motion**

Covers: the two particles

Ask yourself How high does each particle now climb? When do they next reach V? What happens at that collision, and at every later one?

What's really going on

Why the cycloid?

A wheel of radius R rolling under a line traces the cycloid $x = R(t + \sin t)$, $y = R(1 - \cos t)$. Then $\frac{dx}{dt} = R(1 + \cos t)$ and $\frac{dy}{dt} = R \sin t$, so the speed along the curve is $R\sqrt{2 + 2\cos t} = 2R \cos \frac{t}{2}$. Integrating, the arc length from the bottom is $s = 4R \sin \frac{t}{2}$, while $y = 2R \sin^2 \frac{t}{2}$. Hence $s^2 = 8Ry$: the cycloid is exactly the curve with $s = 2\sqrt{ky}$, with $k = 2R$. Conversely, the requirement that the restoring force be proportional to s forces the curve to be a cycloid.

The brachistochrone

In 1696 Johann Bernoulli challenged the mathematicians of Europe: along what curve does a bead slide from one point to a lower point (not directly below it) in the *shortest* time? The answer, the *brachistochrone*, is again a cycloid. Newton is said to have solved the problem in a single evening after receiving it; he published his solution anonymously, but Bernoulli recognised the author, "as the lion by its claw". The problem is the founding example of the *calculus of variations*, which you met in Section 4.7: finding the curve that minimises a quantity defined by an integral.

Huygens's cycloidal clock

Huygens knew that a simple pendulum keeps imperfect time because its period depends on its amplitude. In his *Horologium Oscillatorium* (1673) he showed how to make the bob move along a cycloid: hang the string between two curved "cheeks" shaped as cycloids, and the string wraps round them in such a way that the bob traces a cycloid itself (the *involute* of a cycloid is another cycloid). In practice the cheeks caused more error through friction than they removed, and clockmakers instead

kept the swings small. But Huygens's analysis invented the theory of evolutes and involutes, and his book was one of the great works of seventeenth-century mechanics.

Try this next

1 **Different heights.** A particle is released from height h in the tautochrone bowl. Show that its speed at V is $\sqrt{2gh}$, as it must be by energy conservation, and check that this agrees with ωs_0.

2 **An ordinary bowl.** In a hemispherical bowl of radius R, compare the time taken to reach the bottom from a small height with the time from the rim. Which is longer? (Use the result of Section 11.3.)

3 **Equal masses.** Repeat the second part of the question with two particles of equal mass m, released from heights h_1 and h_2, and coefficient of restitution e. Do they still always collide at V?

4 **Cycloid arc length.** Show that one arch of a cycloid has length $8R$, and that the area under it is $3\pi R^2$, three times the area of the rolling circle.

Further reading

- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). Includes the tautochrone and the brachistochrone, with their history.
- **E. A. Whitman, "Some historical notes on the cycloid"**, *American Mathematical Monthly* 50 (1943). Why the cycloid was called "the Helen of geometers".
- **Paul Nahin, *When Least Is Best*** (Princeton University Press). The brachistochrone and other optimisation problems, told with history.

The full guided solution is in Section 11.4:
https://beyondstep.keithmcnulty.org/ch11/tautochrone.html